OAK RIDGE National Laboratory | LEADERSHIP COMPUTING FACILITY

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Containers Can Be Useful To You, The HPC User

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NCCS

U.S. DEPARTMENT of ENERGY

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What are containers?

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Just a process!

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Confined environment with all your dependencies
(shared libraries, compilers, /usr, whatever)

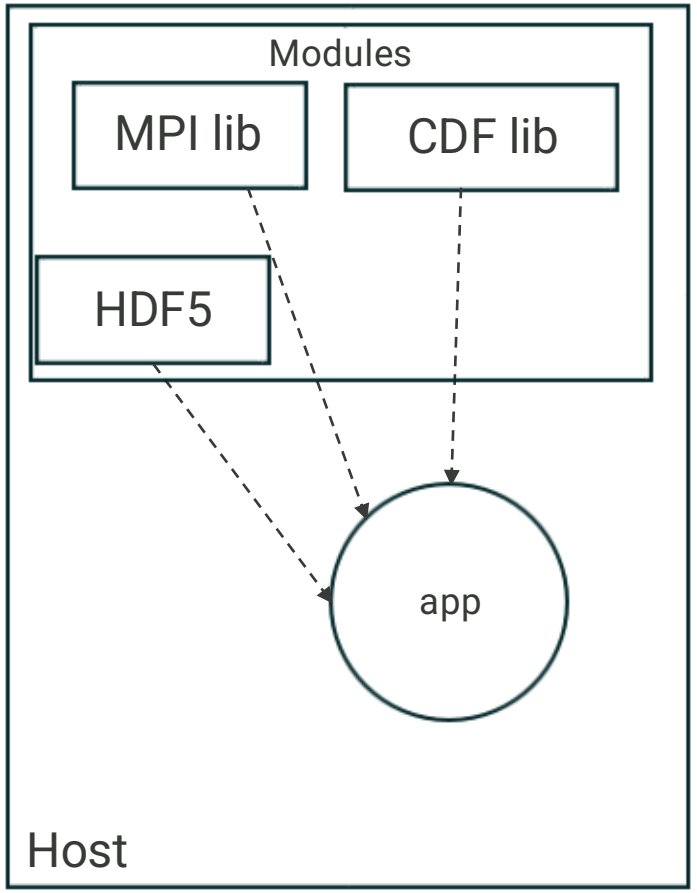
Just a process!

What are containers?

Why you should use containers

- **Containers package your application along with its software environment**
  - Removes dependence on host system's stack
  - Finer control over system level libraries also
  - Still supports optimized host MPI libraries if you need them
  - And no negative impact on performance
- **Byzantine build systems are easier to manage**
- **Potentially easier to port to cloud or other systems***
  - Save time setting up their environment from scratch
- **Fulfill conference software artifact requirements**
- **Easy to iterate and experiment with builds**
- **Take advantage of existing containerized software (like AMD's container releases)**
- **Easier to collaborate**
    - Just hand a container image to a new user on your project
    - build on top of other people containers. Give your container to someone else to build on top of

Containers package your application along with its software environment




```
module load mpich
module load netcdf
module load hdf5
```

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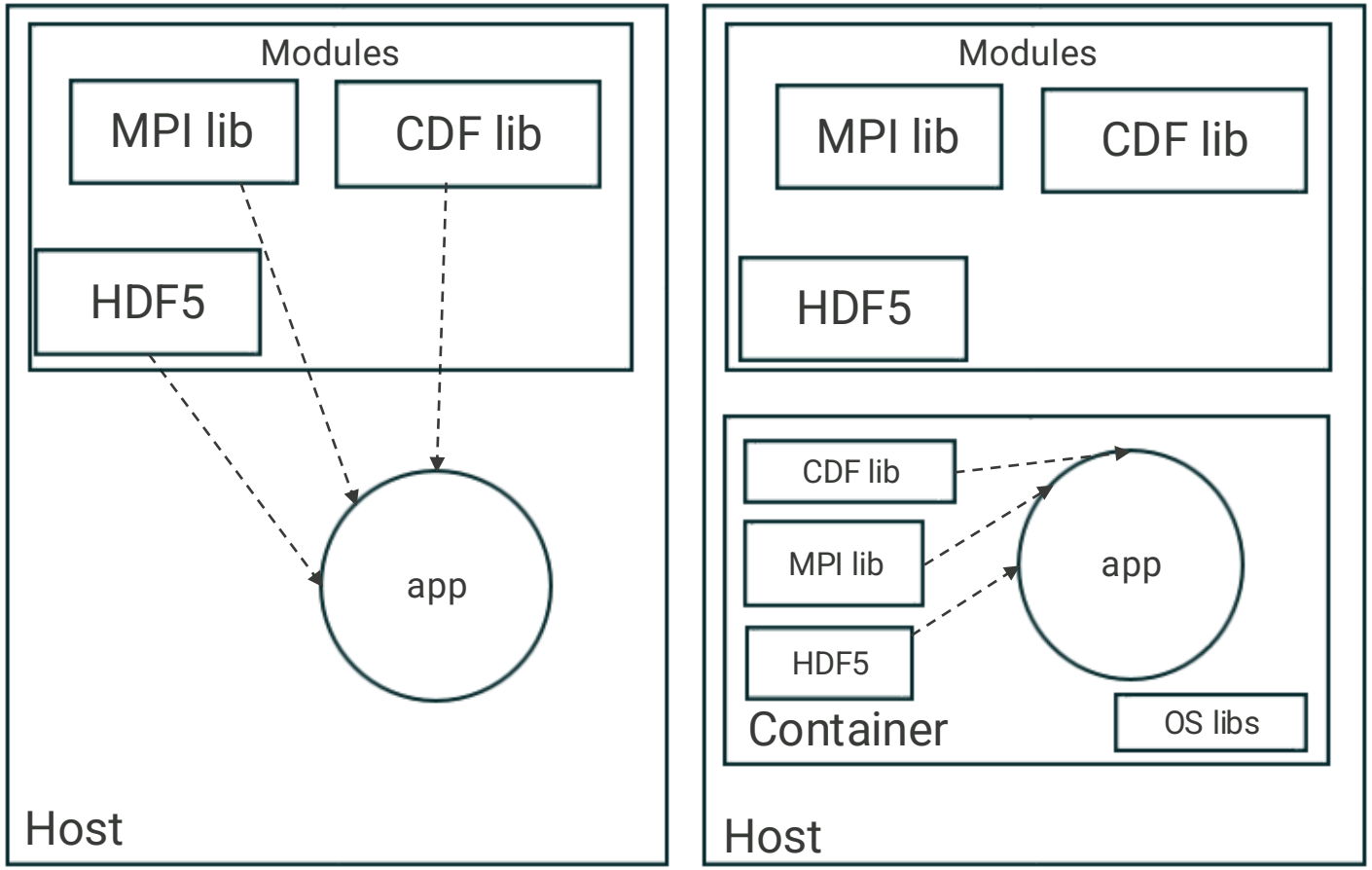

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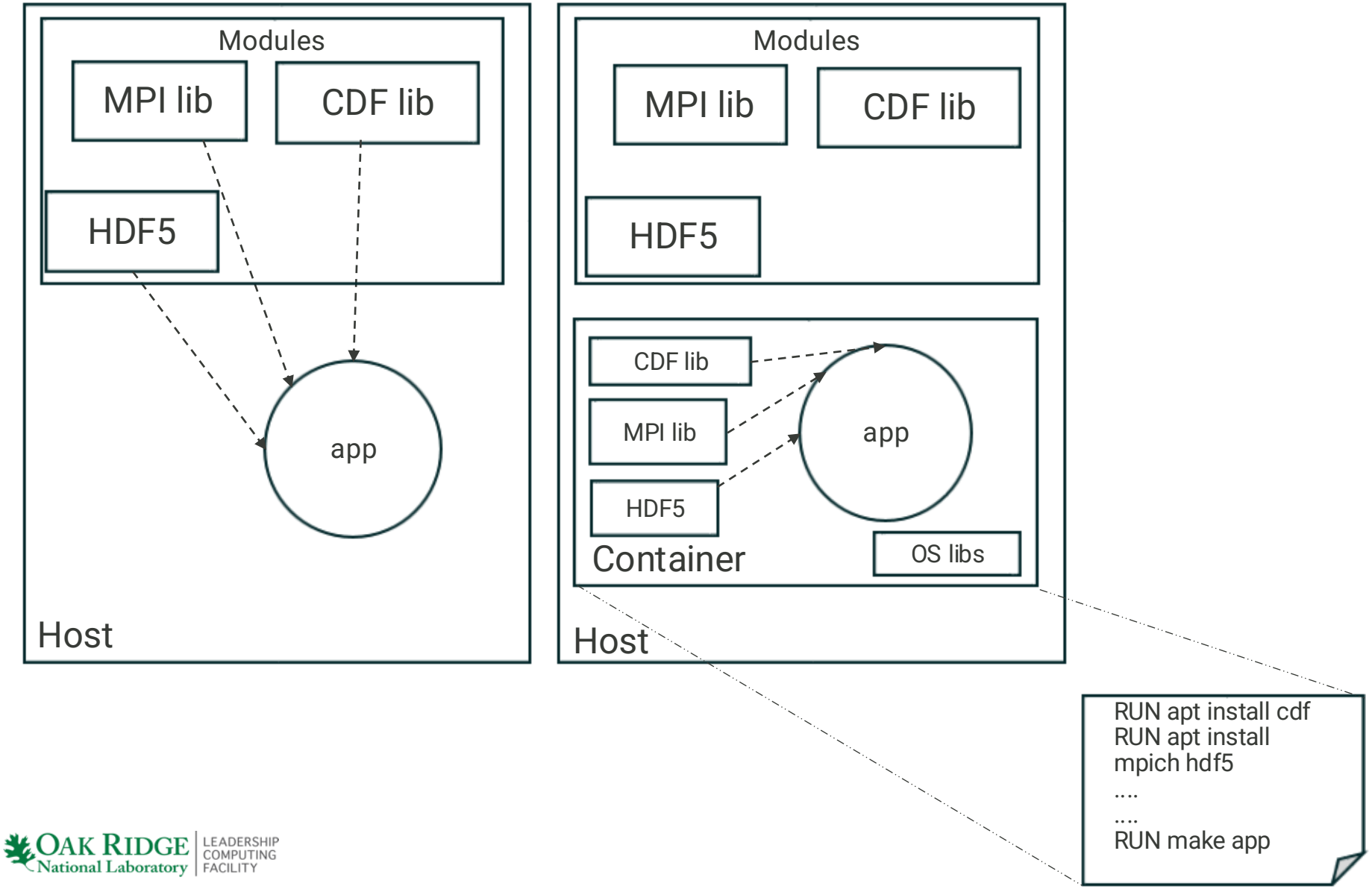

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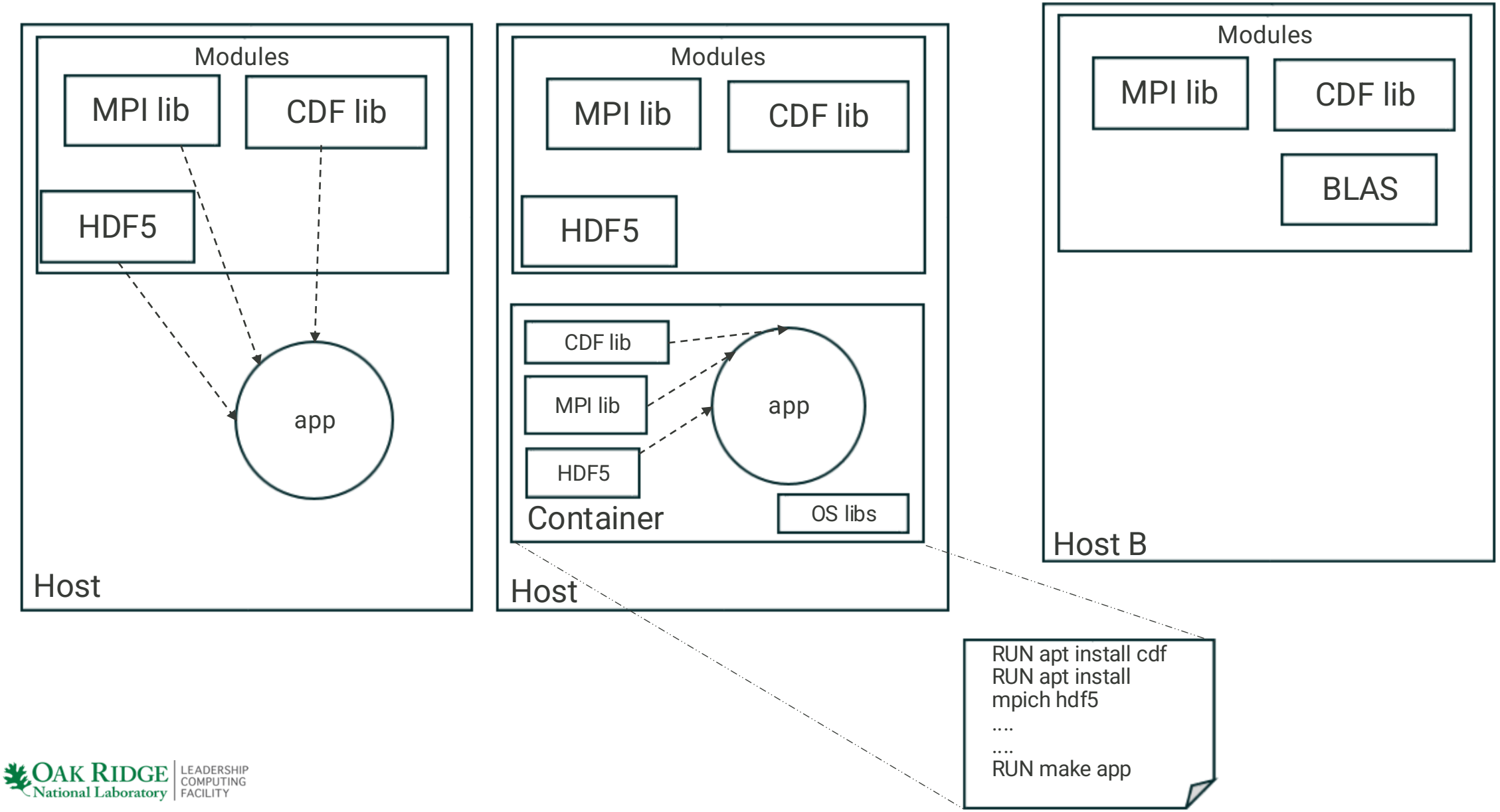

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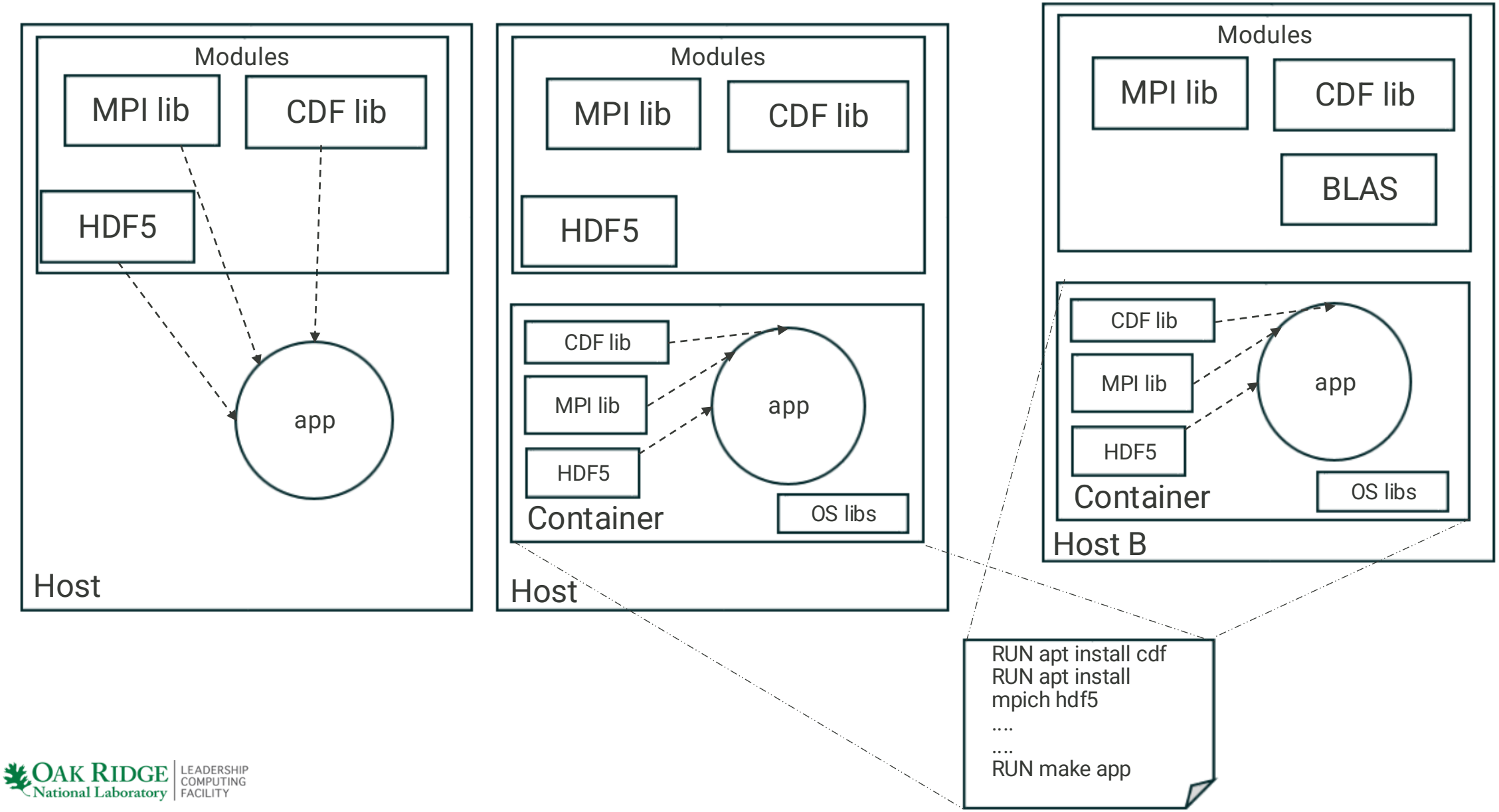

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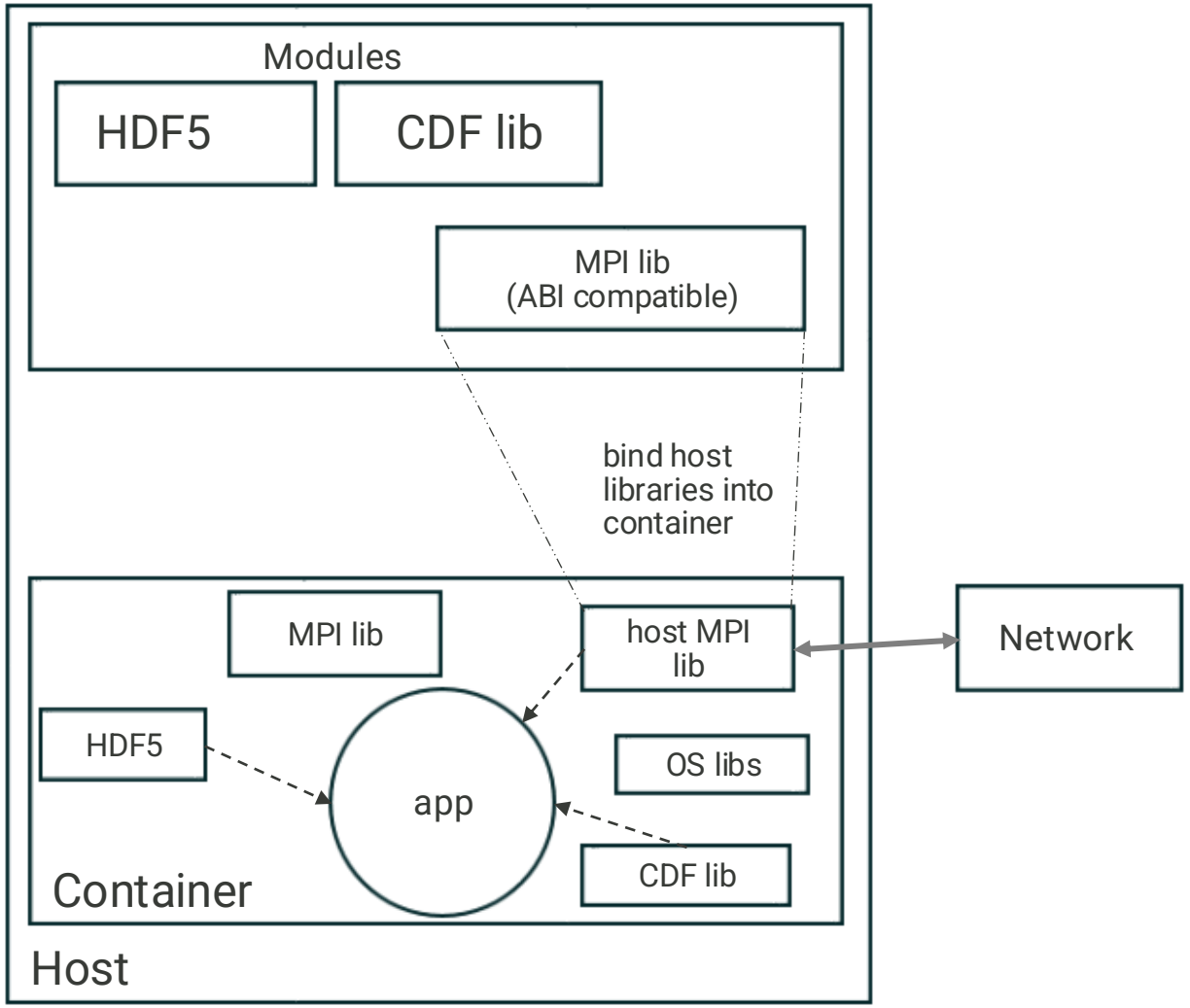

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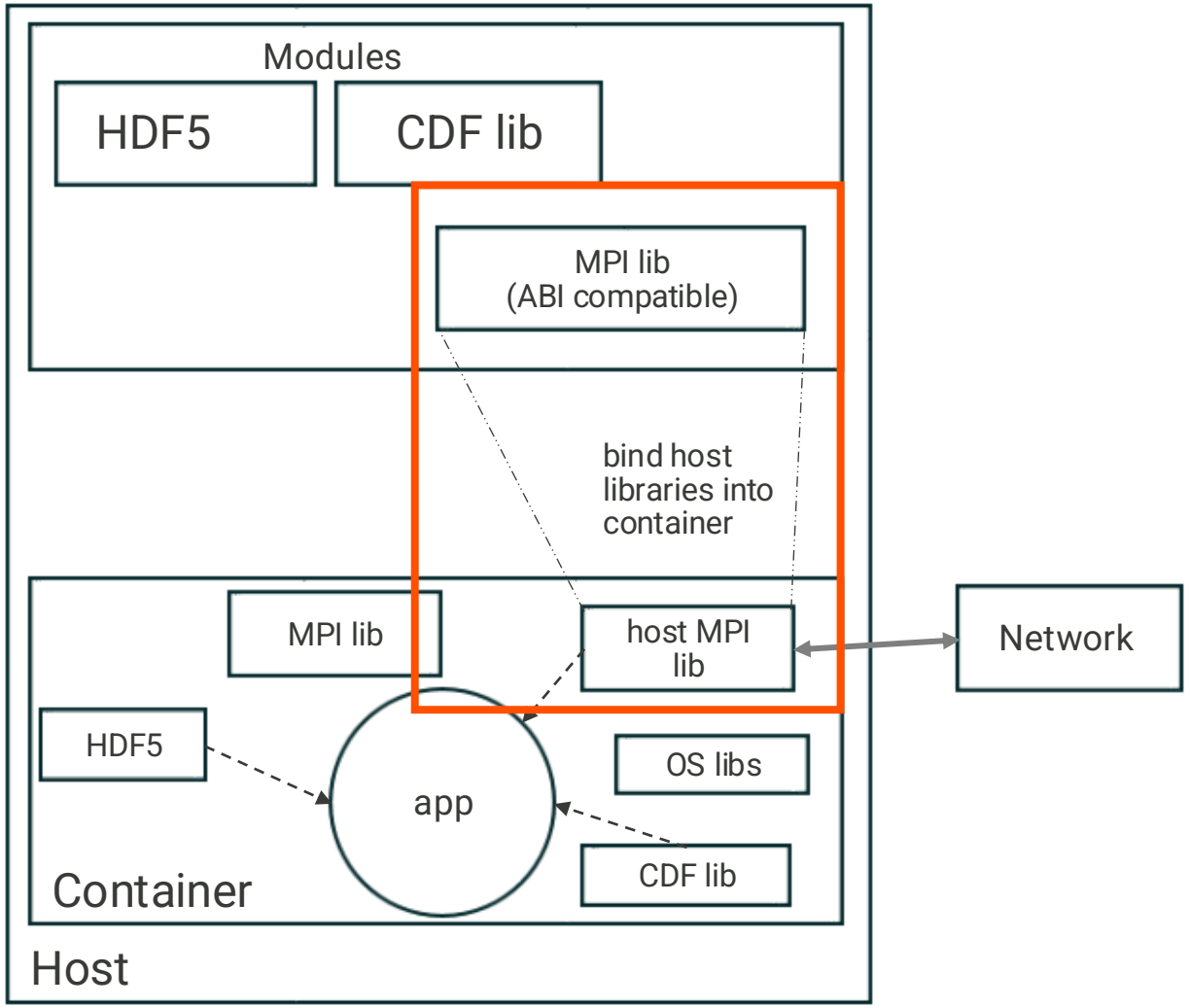

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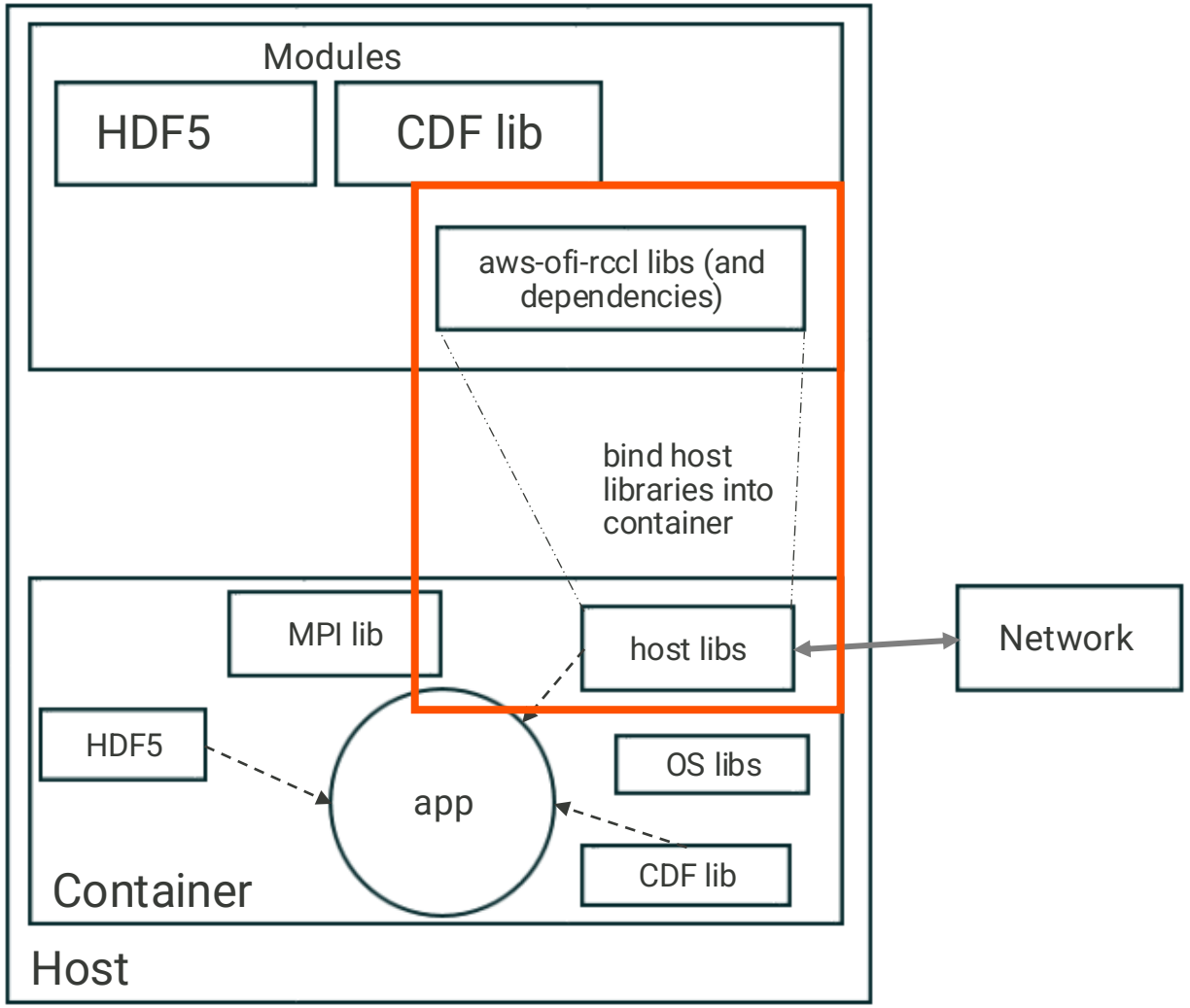
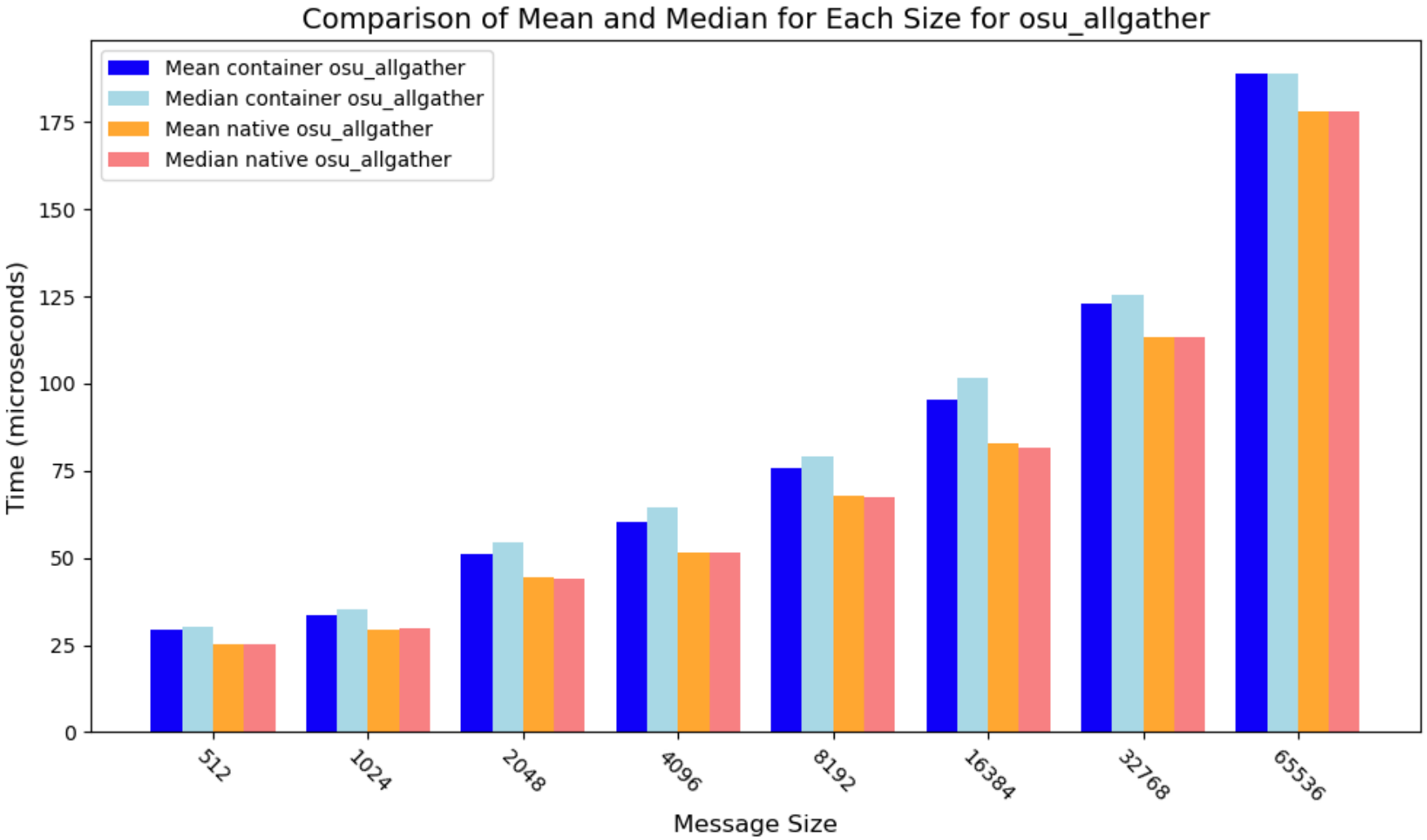
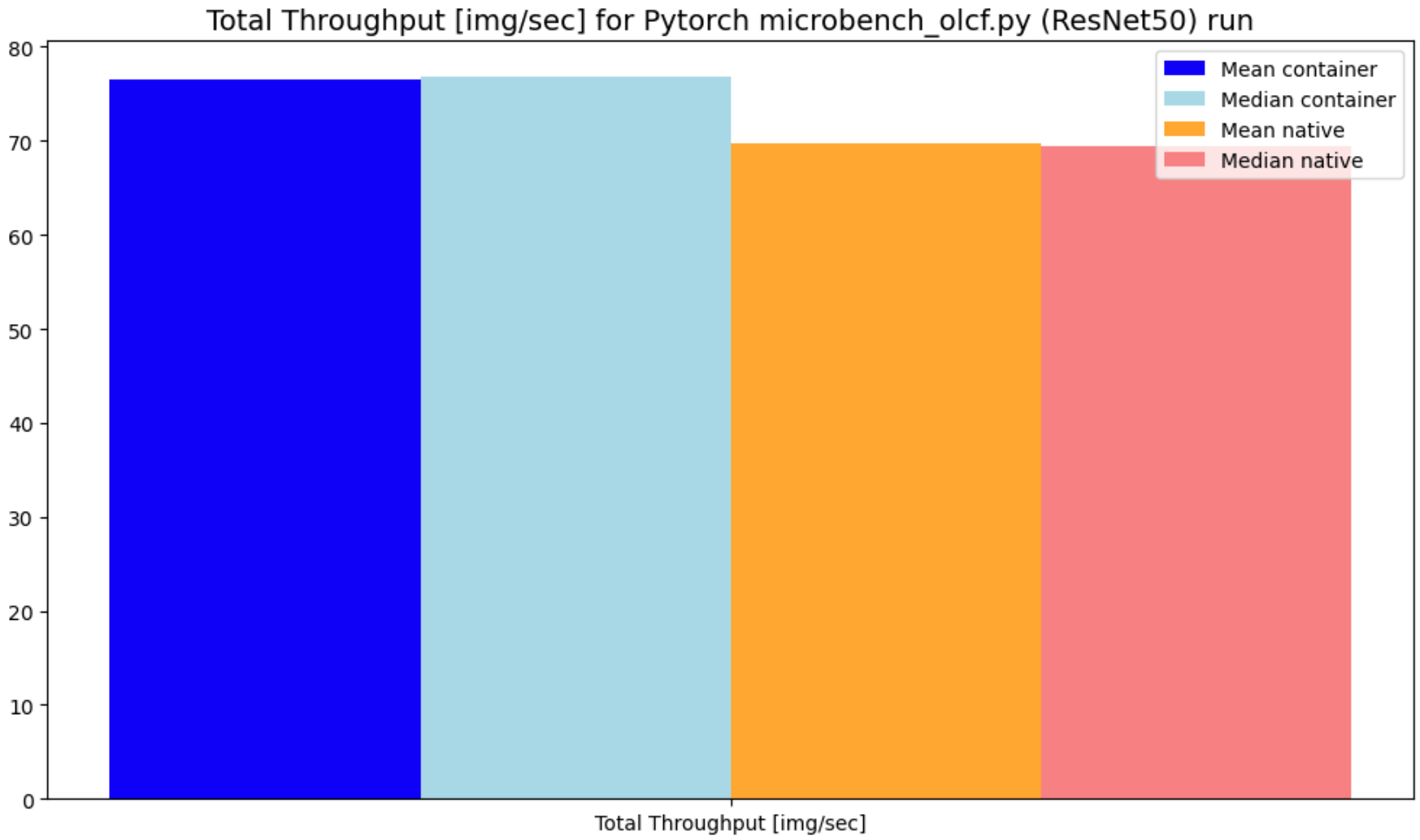


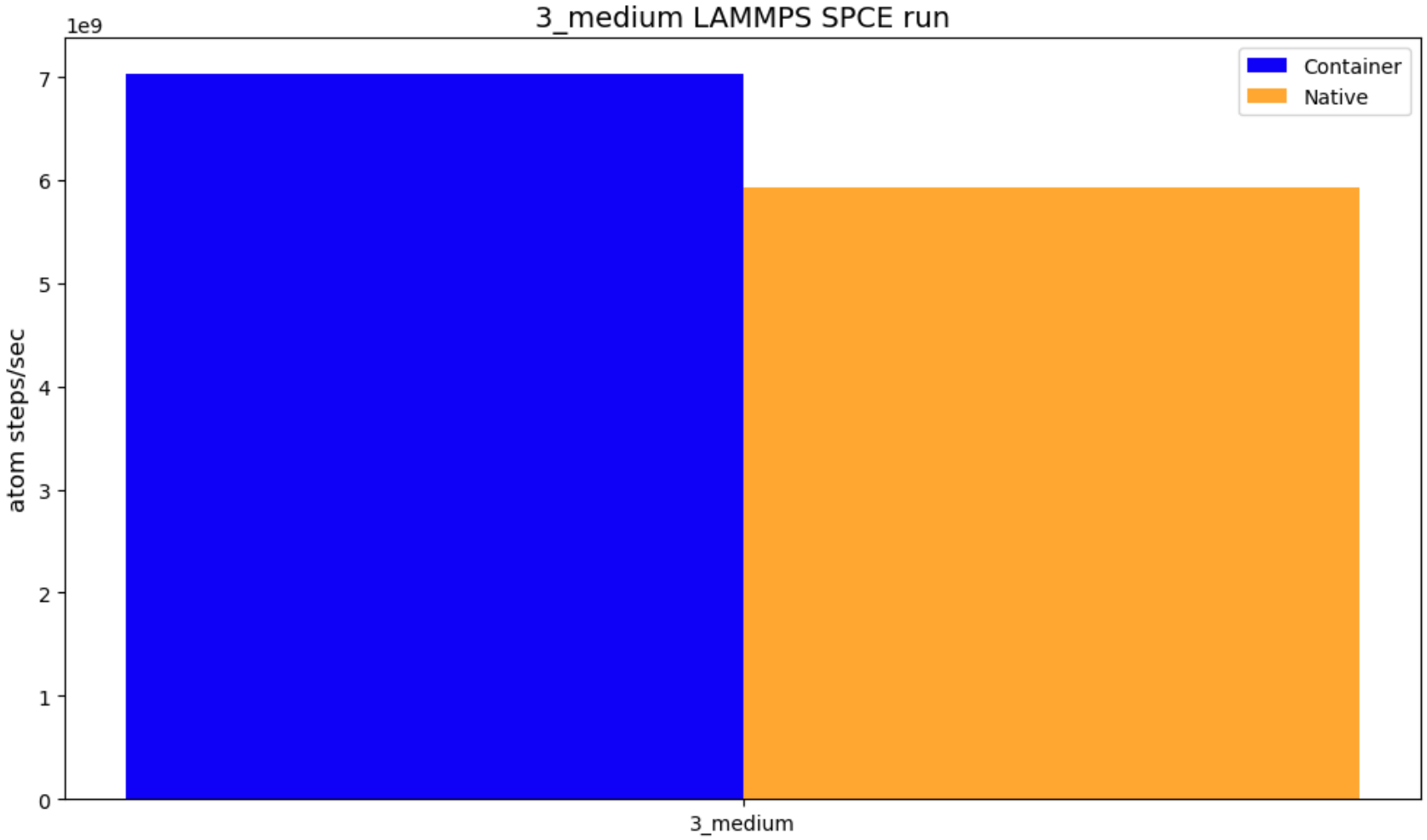
Performance comparison for OSU benchmarks (4 nodes) (lower is better)



Comparsion of throughput for ResNet50 Pytorch benchmark (4 nodes) (higher is better)



Comparison of LAMMPS runs from OLCF6 SPCE benchmark[1] (512 nodes) (higher is better)



[1]: https://www.olcf.ornl.gov/draft-olcf-6-technical-requirements/benchmarks/

OLCF Support for Containers

- Has been present since Summit
- We use Apptainer primarily
- Documentation: https://docs.olcf.ornl.gov/software/containers_on_frontier.html
- Live support during Office Hours: (https://my.olcf.ornl.gov/office_hour_user/scheduler)
- Help desk email: help@olcf.ornl.gov
- Podman for build support in the works
- Would love to provide one-on-one assistance to anyone that wants to try containers!
- Would love to run scaling tests, find bugs, and squash them

Useful Links

- Frontier container docs: https://docs.olcf.ornl.gov/software/containers_on_frontier.html
- Frontier container examples: https://github.com/olcf/olcf_containers_examples/tree/main/frontier
- Apptainer docs: https://apptainer.org/docs/user/main/index.html
- AMD ROCm Dockerhub page: https://hub.docker.com/u/rocm
- Repo for the benchmarks in this presentation: https://gitlab.ccs.ornl.gov/subil/scieng_container_benchmarks#